



Effect of hypersonic flow chemical composition on the oxidation behavior of a super-strong UHTC

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ABSTRACT

A super-strong ZrB₂ ceramic containing WC and SiC was tested in a supersonic arc-jet wind tunnel by exposure to flows with two chemical compositions, simulated air or pure nitrogen, at temperatures of 2650 and 2800 K. Temperature jumps of 500–600 K were observed in both environments at constant flow conditions. SEM analyses revealed that oxygen in the high-enthalpy flow retards the material consumption owing to the formation of partially protective glass. Then, it appears that even 5 vol% of W-compounds is sufficient to modify the subscale oxide configuration and form a cell-like multi-layered architecture typical of tungsten, rather than that typical of ZrB₂.

1. Introduction

Substantial advancements in hypersonic flight technology and development of new low-cost space launch systems are possible only if new materials able to meet the requirements of such demanding thermal loadings are developed. Hypersonic flight vehicles with velocities above Mach 5 experience extremely high aerodynamic and thermo-chemical loads that can seriously impact vehicle structural integrity. Surface temperatures exceed 2000 °C so that the primary requirements of new materials are high temperature strength and oxidation/ablation resistance [1,2]. Ultra-high temperature ceramics, and zirconium diboride (ZrB₂) in particular, are candidate materials for hypersonic environments due to melting points that exceed 3000 °C and exhibit excellent ablation resistance [1].

A ceramic composition system that has attracted notable attention in the last 10 years includes ZrB₂, SiC, and WC [3–14]. This system, where SiC and WC phases vary in amount from 3 to 20 vol%, has been thoroughly studied from the perspective of the densification, mechanisms of microstructure development, oxidation, mechanical properties and very recently, also from the thermal and electrical points of view. Silicon carbide is typically added to enhance the oxidation behavior up to 1700 °C [15–19]. As for WC, it has been ascertained that it has multiple functions, starting from oxide removal during the densification process [4,5], to boron oxide stabilization upon oxidation [20] and to improvement of the mechanical strength at high temperatures [8,21]

owing to the development of a peculiar sub-structure called “core-shell”, where the core is ZrB₂ and the shell a (Zr,W)B₂ solid solution. In this respect, it has been recently assessed that suitable microstructure tailoring and exploitation of the core-shell structure can lead to materials possessing strength above 900 MPa at 1800 °C and remain over 600 at 2100 °C [21], due to in situ grain refinement and local toughening.

On the other hand, a low oxidation rate is a required property in materials envisaged for hot structural applications. Enhanced oxidation behavior have been related to partial consumption of the incoming oxygen to form an adherent oxide layer composed of either a dense outermost scale and/or a compact and stable sub-scale. In the ceramic at hand, SiC forms an outermost SiO₂ glassy coating and WC increases the glass stability and forms solid compounds in the subsurface at low oxygen partial pressure [22]. First studies on the oxidation behavior of this system mostly dealt with static furnaces, up to 1650 °C. The oxidation mechanism has been disclosed upon thorough investigations of the microstructure evolution by scanning and transmission electron microscopy, SEM and TEM [6]. Those analyses first pointed out the unique oxidation mechanism of the (Zr,W)B₂ solid solution into ZrO₂ grains encasing nano-beads of metallic W. In addition, a recent study on the ablation behavior upon exposure to an oxyacetylene flame at 2670 K [14] assessed the effect of W-compounds in suppressing the active oxidation of SiC through their preferential oxidation. As a matter of fact, the combination of excellent high temperature mechanical and

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thermal properties of $\text{ZrB}_2\text{-SiC-WC}$ makes it a robust candidate as an ablation resistant material for components of hypersonic vehicles.

As suggested in [23], in order to properly characterize such materials for the final application, the experimental environment needs to simultaneously reproduce heat flux, gas composition, surface pressure, and flow velocity typical of atmospheric re-entry, which are reliably realizable only in hypersonic arc-jet wind tunnels [24–27]. In this way, it is possible to assess if the beneficial effect of WC addition to $\text{ZrB}_2\text{-SiC}$ UHTCs, demonstrated in furnace environment [6], is confirmed in relevant hypersonic flight conditions.

In particular, although it is well known that the oxidation behavior of ultra-high-temperature ceramic material is crucial for their application in harsh environments, only limited data are available in reduced oxygen partial pressure environments and supersonic flows, which are the most relevant conditions for the application of these materials as leading edges for hypersonic vehicles, subject to extreme heating in the upper atmosphere during re-entry. When the oxygen partial pressure is high enough and a considerable surface layer of protective borosilicate glass is formed, this can provide a reduction of the oxidation rate, preventing, as a barrier, oxygen diffusion towards the bulk material. On the other hand, when oxygen partial pressure is ultra-low, different mechanisms may occur.

The objective of this study is thus to investigate the oxidation behavior of a very high strength ZrB_2 -based ceramic under high-enthalpy supersonic flow with two different chemical compositions. Therefore, two $\text{ZrB}_2\text{-SiC-WC}$ coupons were exposed to an arc-jet flow composed of simulated air, with oxygen partial pressure in the order of thousands of Pa, or pure nitrogen, with oxygen residual partial pressure two orders of magnitude lower, achieving temperatures up to 2650 and 2800 K, respectively. Microstructural analysis enabled to propose the mechanism underpinning the oxidation behavior of this system in extreme environment containing diverse fractions of oxygen.

2. Experimental

2.1. Material

A ZrB_2 - 5 vol% SiC - 5 vol% WC ceramic pellet was sintered to full density according to the methodology described in a previous work [5].

From the sintered disc, two flat button-shaped coupons were machined with a 5 mm-thick head, 17 mm diameter and an overall length of 10 mm including the cylindrical rear pin (for nominal dimensions, see Fig. 1a). Both samples had sharp edges on the front and rear surface. The geometry chosen is that typically used in the available arc-jet

facility [28].

2.2. Arc-jet apparatus

A supersonic continuous, open-circuit arc-jet wind tunnel was employed for the experimental characterization of the UHTC samples. A specified mass flow rate of nitrogen flows through an industrial torch with maximum power of 80 kW, where an electric arc generates plasma. The torch is connected downstream to a mixing chamber, where, if needed, cold oxygen can be injected. The mixture then passes through a convergent-divergent conical nozzle and is accelerated to a nominal Mach number around 3 and exhausted into the test chamber, where pressure is lowered by two vacuum pumps in series. The torch, the mixing chamber and the nozzle are cooled by water flowing in a double wall. A global energy balance, based on the procedure described in [29], is used to evaluate the average specific total enthalpy at the exit of the torch and of the nozzle (H_0). At a gas mass flow rate of 1 g/s, the stagnation-point pressure generally assumes values in the range 6–12 kPa at different levels of torch power [2,30,31].

Material samples were placed inside the wind tunnel test chamber, downstream the nozzle, at a distance of 1 cm from its outlet section, in order to be exposed to the high-enthalpy supersonic flow.

The surface temperature of the samples was measured ($\pm 1\%$ instrumental accuracy) by means of a digital pyrometer (Infratherm ISQ5, Impac Electronic GmbH, Germany) at an acquisition rate of 100 Hz, operating, in the range 1273–3273 K, in the two-color mode (the used wavelength bands are 0.7–1.15 μm and 0.97–1.15 μm). Moreover, an infrared (IR) thermo-camera (TC, Pyroview 512 N, DIAS Infrared GmbH, Germany) provides the temperature distribution on the material surface, in the range 873–3273 K ($\pm 1\%$ of measured value in K at temperatures lower than 1673 K; $\pm 2\%$ at temperatures higher than 1673 K), operating in the spectral range from 0.8 to 1.1 μm . The procedure employed to set the correct value of spectral emittance, on which the IR-TC measurement is dependent, is reported in [32].

High-Definition videos of the tests were recorded by means of a Camera Flea3 1.3 MP Color USB3 Vision with a resolution of 1328×1048 and a frame rate equal to 25 fps.

2.3. Characterization

Crystalline phases formed after the arc-jet tests were identified by X-ray diffraction (XRD, mod. D8 Advance - Bruker, Germany).

The microstructure before and after torch tests was analyzed on the surface and cross section by scanning electron microscopy (FE-SEM,

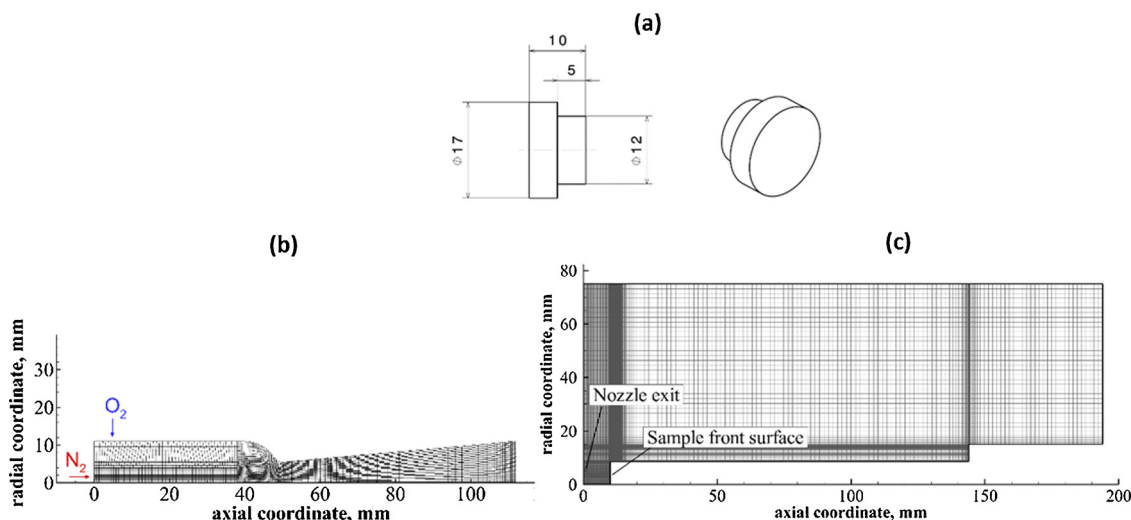


Fig. 1. a) Nominal design of UHTC specimens. Computational domains for the CFD simulation of the flow field: b) mixing chamber and nozzle of the arc-jet facility and c) test chamber.

Carl Zeiss Sigma NTS GmbH, Oberkochen, DE) and energy dispersive x-ray spectroscopy (EDS, INCA Energy 300, Oxford instruments, UK).

2.4. Numerical models

Computational Fluid Dynamics (CFD) simulations were performed by means of a commercial solver, with the aim of reconstructing the fluid field and evaluating the aero-dynamic and thermal loads on the material samples.

The steady Reynolds-Averaged Navier-Stokes equations are solved for a turbulent multi-reacting gas mixture with a control-volume-based technique and a density-based algorithm. A finite rate reaction model, with rate constants calculated by Arrhenius law, is used to describe the evolution of a gas made of up to five species (O, O₂, NO, N and N₂). The transport properties of the gas species are obtained by the kinetic theory of gases. For the calculation of density, mixture is modelled as an ideal gas and a mass-weighted mixing law is considered for the mixture transport properties.

The computations are carried out dividing the 2D-axisymmetric domain into two separate parts: mixing-chamber and nozzle (Fig. 1b) and test chamber (Fig. 1c).

The hot gas (N₂, red in Fig. 1b) coming from the torch is injected into the domain of Fig. 1b through an axial mass flow inlet, representing the exit section of the torch. Total temperature and gas composition, corresponding to the specific total enthalpy measured at torch exit, are calculated by means of the NASA CEA (Chemical Equilibrium with Applications) software [33]. In case of air flow, the cold gas (O₂, blue in Fig. 1b) is injected in the mixing chamber through a radial mass flow inlet. The top surface of the domain is treated as a no-slip stationary wall at temperature $T = 400$ K (in order to simulate water-cooling of the nozzle). The nozzle exit section is modelled as a supersonic pressure outlet.

The conditions achieved at nozzle exit are then used as input for the simulation of the flow field around the sample, in the computational domain of Fig. 1b. Here, the nozzle exit section is described as a velocity inlet. On the top and rear boundaries of the domain, a pressure-outlet condition corresponding to the experimental vacuum environment is set.

The front surface of the sample is treated as a cold no-slip wall for the evaluation of the aero-thermo-dynamic loads. The effect of surface catalycity, which is the ability of the material surface to promote exothermic recombination reactions at the wall, is also taken into account. Two limit cases can be considered: non-catalytic wall, assigning zero species diffusive flux on the wall, and fully catalytic wall, assigning zero atomic species concentration at the wall.

2.4.1. Thermal analysis of the solid components

Besides the simulation of the fluid field, also a steady-state thermal analysis of the solid material has been performed, solving by a volume-control technique the energy equation in an axisymmetric computational domain including the sample and its supporting elements. The aim was essentially to provide an estimation of the material thermal conductivity.

During exposure to high-enthalpy flow, the convective heat flux entering the samples surfaces (including the chemical contribution) is partly transferred away by conduction through the solid and partly re-radiated back to the environment. At steady state, global radiative equilibrium is achieved among the different heat flux components. Hence, in the simulation, the sample front surface, directly exposed to the hypersonic flow, is modelled as a wall with assigned (conduction) heat flux, whereas on the other solid edges an external radiation condition is set, assuming an external temperature of 400 K. The value of the total surface emissivity is assumed equal to the spectral emissivity derived from the IR-TC measurements, since at ultra-high temperatures most of the power is irradiated in the IR-TC operating wavelength band [34].

For each simulation, the axial temperature trend on the side edge of the sample is compared with the experimental profile provided by the thermo-camera. The values of the conductive heat flux assigned on the front surface and of the UHTC material thermal conductivity were changed parametrically in each simulation, until a good match between experimental and numerical results was achieved, with thermal conductivity mainly influencing the temperature axial profile.

3. Results

3.1. Arc-jet tests

Tests were carried out considering two different atmospheric compositions: simulated air, composed by 80% nitrogen and 20% oxygen, and pure nitrogen. The estimated oxygen partial pressure for the first case is in the order of 1000–2000 Pa (about 20% of the total stagnation point pressure calculated by numerical simulations, see later Fig. 17c), whereas, when the arc-jet flow is of pure nitrogen, considering the vacuum level in the test chamber, oxygen partial pressure is two orders of magnitude lower. In both cases, the total gas mass flow rate was 1 g/s. The torch arc power, and consequently the flow specific total enthalpy, were increased stepwise. The enthalpy level and the duration of each step for the two tests is reported in Tables 1 and 2, respectively. After the maximum enthalpy step, the power was decreased gradually until facility shutdown.

Fig. 2 shows the temperature histories of the two samples, as detected by the pyrometer. It can be noticed that the thermal histories are comparable at relatively low enthalpies, then they tend to diverge. Both samples experienced a rapid rise in temperature at constant flow enthalpy, which herein will be referred to as “temperature jump”. The behaviour of the two samples is different though. In particular, the sample tested in nitrogen atmosphere experienced a sudden jump at constant enthalpy (about 18 MJ/kg, step 5n) when the temperature approached 2000 K, and its final temperature exceeded 2800 K. The sample tested in air had instead a more gradual jump, during the last step (enthalpy of around 20 MJ/kg, step 7a), and reached temperatures over 2600 K only at the end of the test, which was stopped about 30 s after the occurrence of the jump.

The temperature jump has been associated with a transition in surface chemistry that leads to the loss of protective silica glass and substantially increases the chemical component of the heat flux delivered to the surface, besides dramatically reducing the thermal conductivity in the oxide scale [28,35].

The surface spectral emissivity at the wavelengths around 1 μ m was estimated by comparison between pyrometer and thermo-camera measurements, and it was observed to be about 0.85 at all temperatures.

Fig. 3 shows some thermal images of the two samples taken by the DIAS Pyroview 512 N thermo-camera at different time instants, during test. Fig. 3a refers to sample 1, tested in air. The picture on the left shows the sample during step 7a, before the occurrence of the temperature jump. In this phase, the temperature of the sample is rather uniform, around 2100–2200 K. The temperature jump appears then to

Table 1
Test conditions in simulated air flow.

Step	Specific total enthalpy [MJ/kg]	Step Duration [s]
1a	8.5	30
2a	10.0	30
3a	12.0	30
4a	13.5	30
5a	15.5	30
6a	18.0	60
7a	20.0	105

Table 2
Test conditions in nitrogen flow.

Step	Specific total enthalpy [MJ/kg]	Step Duration [s]
1n	8.0	30
2n	10.5	30
3n	13.0	30
4n	15.5	30
5n	18.0	60
6n	21.0	120

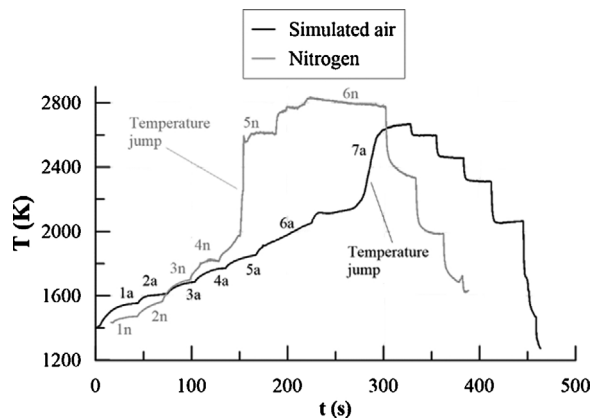


Fig. 2. Surface temperatures measured by pyrometer IMPAC ISQ5 on the two samples.

be a dynamic process, leading to the formation of an ultra-hot region in the periphery of the exposed surface of the sample, which extends its size in time, as observable by comparison between the central and the right picture, while the rest of the sample remains at a much lower temperature.

Fig. 3b shows instead the thermal evolution of the sample tested in pure nitrogen. Just before the temperature jump (picture on the left), the temperature is uniform, around 2000 K, and bubbling is observable on the front surface of the sample. During and after the sudden jump, bursting of localized bubbles occurs, resulting in hot spots dispersed all over the surface (central picture). At the end of the heating phase, the surface reaches a uniform ultra-high temperature, whereas the rear region is kept at lower temperatures.

The observed phenomena occurring during exposure to arc-jet supersonic flow are also highlighted by captions of the video recorded during test by the Flea3 optical camera, shown in **Fig. 4**. We can observe that when the impinging flux includes oxygen, **Fig. 4a**, initial outgassing of vapor fumes occurs. The blueish color suggests it to be based mainly on W-O [36]. Then asymmetric heating on the button crown occurs and, as soon as the circle is completed, formation of borosilicate glassy-based drops/bubbles at the side edges is visible and remains with a dark contrast during cooling. As for the test in nitrogen, **Fig. 4b**, the formation of hot tiny bubbles is limited to isolated spots homogeneously spread on the surface, then they coalesce, the lateral sides are heated until the bottom starts to bubble as well. The last two pictures of **Fig. 4b** correspond to instants of advanced sample cooling, and therefore only very thin oxide scale is still visible because it is the only hot region. However, it can be appreciated that the largest bubble on the top surface breaks upon cooling, owing to tensile stresses rising in the oxide scale.

The microstructure modifications occurred after the tests are proved by the white color assumed by the samples heads, **Fig. 5**. It is visible

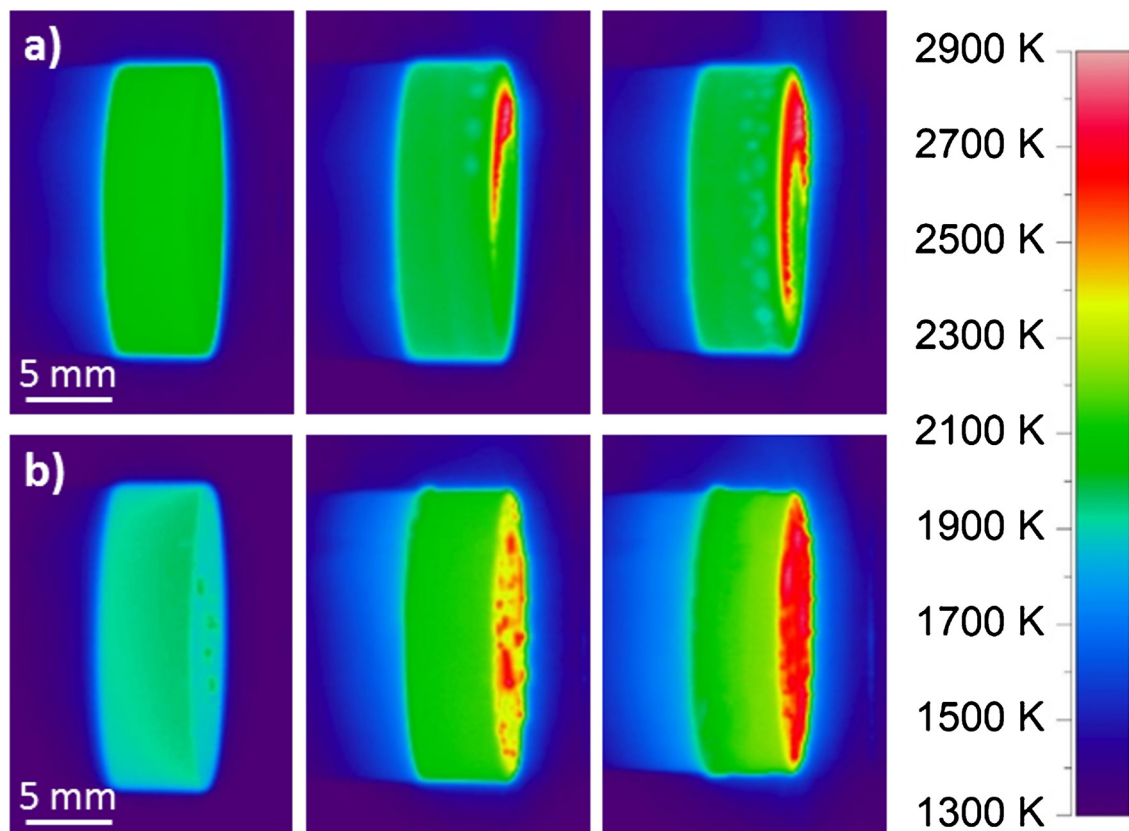


Fig. 3. Thermal images of a) sample tested in air and b) sample tested in nitrogen, just before temperature jump (left), just after temperature jump (center) and at the end of heating phase (right). The temperature jump occurs at $H_0 = 20$ MJ/kg in air (a) and 18 MJ/kg in nitrogen (b).

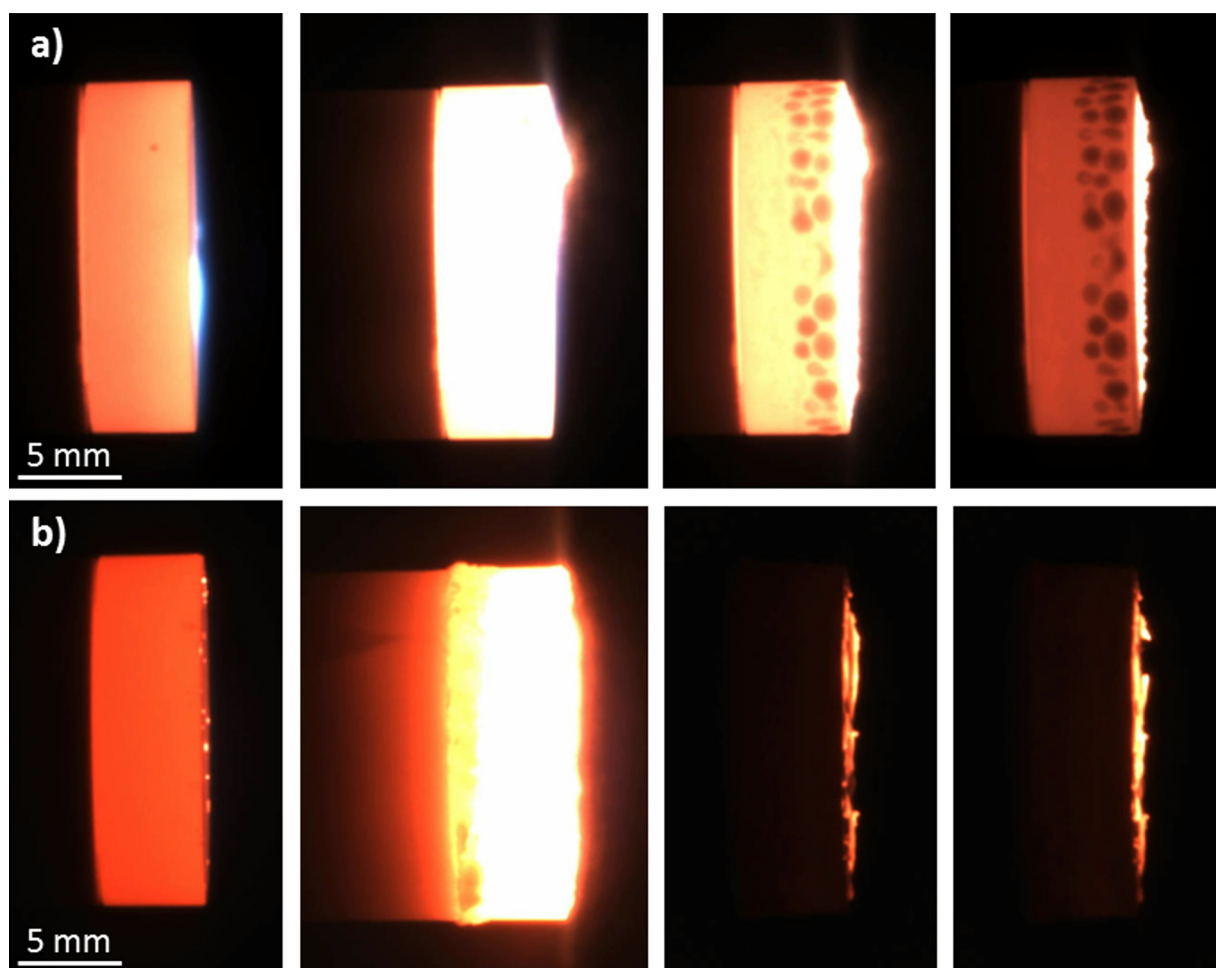


Fig. 4. Frame shots of the video recorded by the Flea3 Camera during a) test in simulated air and b) in nitrogen.

that the formed oxide was not smooth and compact, but rather tended to spall off after both tests, as described in the next sections. Particularly, notable bubbles formation and oxide uplifting are visible in the periphery of sample 1, whilst evidence of bursting phenomena is visible

on all the surface of sample 2.

Despite the evident damages subsequent to the exposure to high-enthalpy flow, the mass is substantially unaltered after test, with a slight increase, possibly due to oxide formation (Table 3).

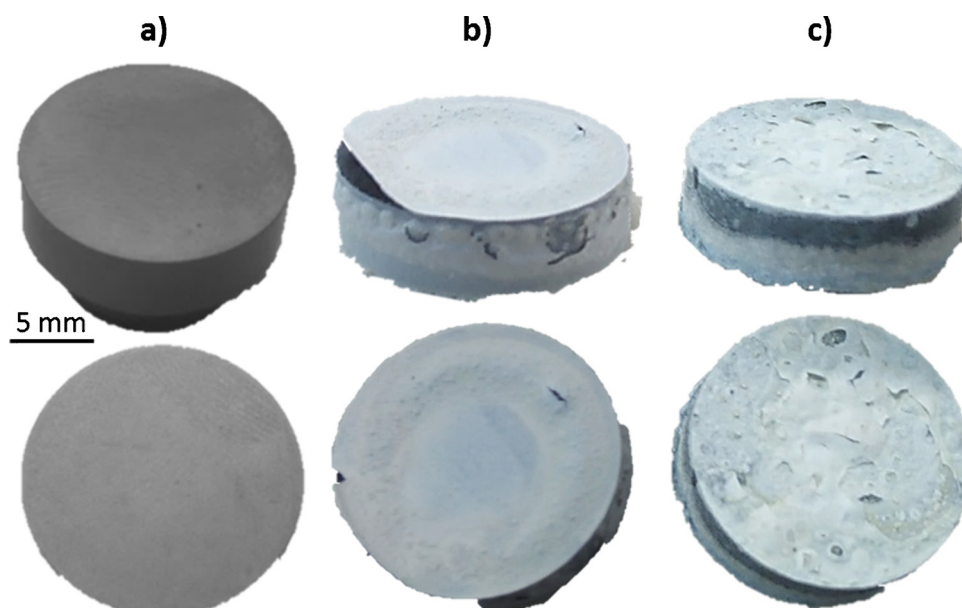


Fig. 5. Pictures of the samples (a) before the tests, (b) after test in simulated air, (c) after test in nitrogen.

Table 3
Samples mass before and after test.

Sample	Initial mass (g)	Final mass (g)
1 (Air)	10.592	10.616
2 (Nitrogen)	10.679	10.697

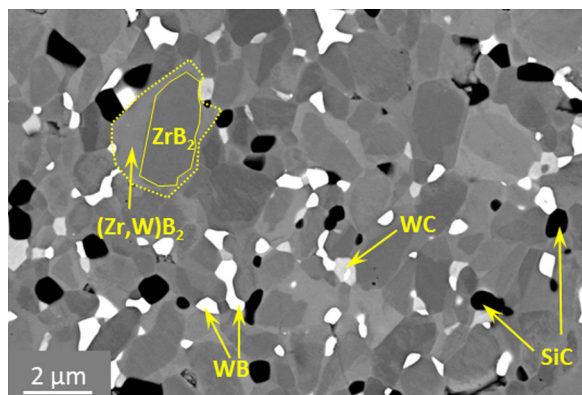


Fig. 6. SEM image of the polished surface of the pristine microstructure evidencing the core-shell structure of the boride matrix and the secondary phases.

3.2. Microstructure evolution

3.2.1. Pristine microstructure

The microstructure of this ceramic has been extensively studied and described in previous papers [5,21]. To briefly summarize the main aspects and to understand the microstructure variation upon the arc-jet flow tests, the boride matrix had mean grain size around $1.8 \pm 0.6 \mu\text{m}$ displaying the typical core-shell substructure, with ZrB_2 being the core and a $(\text{Zr,W})\text{B}_2$ solid solution the shell. The secondary phases were homogeneously distributed: SiC is recognizable with dark contrast in Fig. 6, WB has white aspect, whilst residual WC is light grey. The formation of the WB phase is due to the chemical reaction between the initial WC and B_2O_3 covering the diboride powder particles [4]. The amount of these secondary phases was estimated by image analysis to be ~5 vol% SiC, about ~4.9 vol% WB and less than 1 vol% of WC.

3.2.2. Microstructure after exposure to supersonic flow

The microstructure evolution of the two samples after exposure to different atmospheres, already introduced at macroscopic level in Fig. 5, is herein discussed.

X-ray diffraction carried out on the buttons head, Fig. S1, revealed a predominance of monoclinic ZrO_2 in both samples with possible B_2O_3 traces in sample 1 exposed to air and WO_3 traces in sample 2 exposed to nitrogen, see the peaks at ~ 31 and $23/33^\circ$, respectively.

3.2.3. Microstructure after exposure to air flow

The external surface of sample 1 after test in air, reported in Fig. 7a, shows three distinct regions: a periphery, whose details are displayed in Fig. 7b, the center, magnified in Fig. 7c, and the sub-surface made visible upon detachment of the outermost scale, Figs. 8–10.

The nominal temperature achieved by this sample was 2650 K. Due to high shear forces that stripped any glass phase off the surface, the edge of the button was composed of only ZrO_2 featured by bubbles, dense grains and micro-cracks, Fig. 7b. In the center, silica was instead still present owing to the lower shear forces experienced. Zirconia islands and WO_x were found immersed in the glass, Fig. 7c. This last phase, recognizable with bright contrast, was dispersed intergranularly in large ZrO_2 aggregates, Fig. 7c1, or as a “cap” when ZrO_2 was found as tiny precipitates into glass, Fig. 7c2. The reason of this peculiar configuration is presumably related to the precipitation of Zr-W-O from

supersaturated zone within the silica-based glass and to the low miscibility of the two corresponding solid oxides. Since the melting point of WO_3 is 1746 K and the measured surface temperature was 2650 K, we assume the bright phase to be rich in tungsten, rather than oxygen.

Images of the microstructure of the sub-scale are depicted in Figs. 8–10 with zone tags relative to Fig. 7a. The subscale is visible due to partial detachment of the surface oxide. Starting from the edges, Fig. 8, ZrO_2 and WO_3 areas with rough aspect were found, all covered by a boro-silicate glass, recognizable as transparent whiskers in Fig. 8d3/4. Moving towards the center, Fig. 9, the morphology of both oxides notably changed: ZrO_2 was in the form of large isolated grains, Fig. 9e1, WO_3 had a cubic or Maltese cross form and stood out of a W phase, Fig. 9e2, as testified by the EDS analysis. Silica-based glass was present as well. Then, in zone (f) of Fig. 7a, the morphology changed again: ZrO_2 spheres were immersed in WO_3 elongated crisp grains, Fig. 10f1, in turn surrounded by a glass mainly composed of boron oxide, like indicated in the EDS spectrum of Fig. 10f2.

Looking at the cross-section analysis, it is noticeable that oxide uplift occurred at the edges of the button, with an oxide layer – bulk material separation distance of $330 \mu\text{m}$, Fig. 11a. This could explain why the center of the button still displays a silica glass coverage on its surface, when normally this is compressed out towards the sample periphery by the gas impact at the stagnation point. The outermost scale, partially detached from the sample, is a dense ZrO_2 layer about $100 \mu\text{m}$ thick with few scattered WO_3 aggregates, Fig. 11b. Then, beyond a large void (filled by the epoxy resin during cross-section preparation, void I), ZrO_2 is covered by white interconnected particles made of WO_3 and metallic W with intergranular silica-based glass, as identified by EDS, Fig. 11b–d. The tungsten accumulation sits on a dense ZrO_2 -W scale, which, according to the Zr-W-O system relationships [37] and microstructure observation, was possibly melted during the test. The composition of such melt is rich in tungsten oxide, above ~60 mol % WO_3 , and therefore its formation and existence is actually possible only in a scale where W-O volatilization is not favorable, i.e. in the subscale at low O_2 partial pressure. Also this dense scale tends to delaminate from the sample core, void II. Analysis of the button in the center confirmed a much thinner modified scale, about $160 \mu\text{m}$, as shown in the EDS elemental mapping in Fig. 11c. In details and in agreement with the surface analysis, a silica glass sheltered a layer made of coarse equiaxed ZrO_2 grains slightly detached from the underneath zone composed of W interconnected particles that were sitting on columnar ZrO_2 grains, Fig. 11d. These in turn were dotted by tiny bright rounded particles, W droplets, and dendritic WO_3 structures. The columnar grains were anchored to a scale of equiaxed and dense ZrO_2 with W particles. Further below, the parent bulk ceramic was found.

3.2.4. Microstructure after exposure to nitrogen flow

SEM images of the external surface of sample 2 after test in nitrogen flow at 2800 K are displayed in Fig. 12a, where three main zones can be identified: a periphery, an intermediate zone and a central zone. The periphery is mostly porous zirconia with scattered WO_3 aggregates, Fig. 12b. The intermediate region starts with $200\text{--}300 \mu\text{m}$ coarse ZrO_2 stones sitting on an irregular and partially cracked more dense ZrO_2 pavement, decorated by excrescences and intergranular WO_3 phase, Fig. 12c. This region is also characterized by rounded craters, as pointed by arrow in Fig. 12a, residual of bubbles bursting. Zirconia grains in this regions have a polyhedral shape and are featured by diffused tiny porosities. Moving to the central zone, dendritic shaped WO_3 is the carpet for rounded ZrO_2 - SiO_2 agglomerates, occasionally wrapped in WO_3 filaments, Fig. 12d. Then, the actual center is made of equiaxed WO_3 and ZrO_2 dense grains, without evident silica-based glass, differently to what observed in sample 1 in Fig. 7a.

The cross section analysis revealed also in this case substantial oxide lifting with multiple delamination and formation of bubbles at various depth, Fig. 13a,b. The maximum height of the bubbles was in this case about $250 \mu\text{m}$, whilst the modified scale in the least damaged zones was

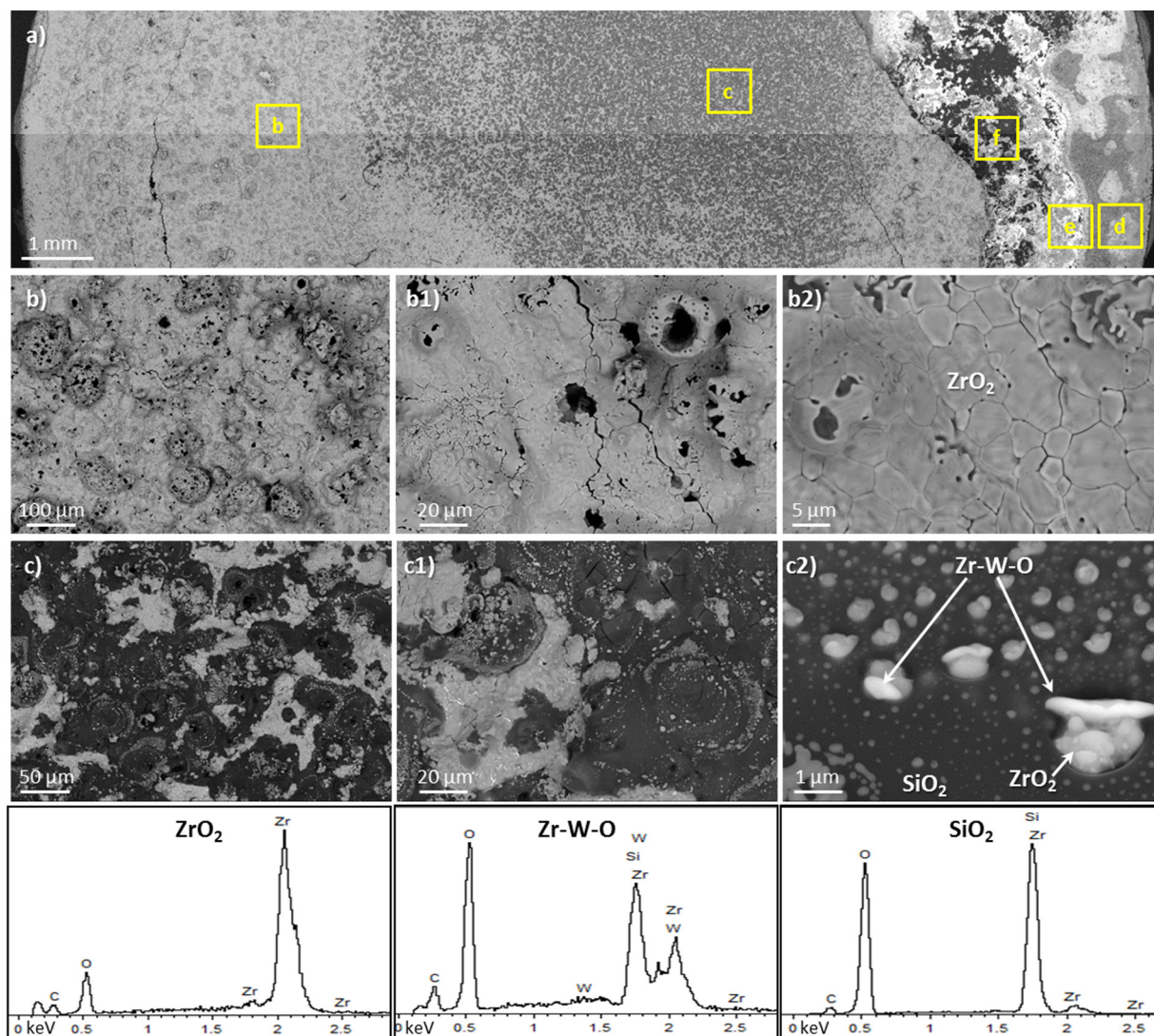


Fig. 7. SEM images of the surface of sample 1 after the test showing a) the overall appearance of the button, b), b1), b2), magnification of the button periphery and c), c1), c2), magnification of the button center. Microstructure details relative to zone d)-f) are depicted in Fig. 10. Below the EDS spectra of the phases tagged in c2).

around 130 μm Fig. 13c,d. The structure of the modified scale was analogous to that described for sample 1: an outermost dense ZrO_2 scale with scattered WO_3 aggregates, large voids, accumulation of WO_3 on dense ZrO_2 with intergranular W particles, then another void was often found before meeting the ZrO_2 rounded grains encasing W nanobeads, Fig. 13e-g. The EDS mapping confirmed this architecture, Fig. 13h.

Two are the main differences with respect to the sample tested in simulated air: first, the outermost scale of the central area was mostly crystalline and based on ZrO_2/W , i.e. without silica-based glass, compare Figs. 7a and 12a. Then, the sub-surface scale of sample 2 was typified by formation of a multi-layered cell-like scale, Fig. 13b, a feature which is just slightly outlined in sample 1, Fig. 11b.

3.3. Numerical results

The models described in Section 2.4 were employed to predict the flow conditions around the test article. The main results are herein summarized.

Fig. 14 shows the distribution of Mach number inside the mixing chamber and supersonic nozzle of the arc-jet facility in the case of air flow and in correspondence of a total enthalpy equal to 20 MJ/kg, Step 7a in Table 1. The slightly supersonic core jet coming from the torch is visible on the axis. Downstream, nitrogen and oxygen mix and the flow

becomes more uniform in the radial direction. As expected, the Mach number at nozzle exit achieves a value close to the nominal one of 3.

Fig. 15 shows a comparison of the thermo-fluid-dynamic conditions and the chemical composition of the two flows around the samples, during the two steps in which the temperature jump happened (step 7a for sample 1, air, and 5n for sample 2, nitrogen). It is possible to see a normal shockwave forming in front of the test article, which makes the pressure rise to levels of almost 9 kPa (P images in Fig. 15a,b). The analysis of N and O mass fraction distribution around the test article (N and O images in Fig. 15a) show that there is a considerable amount of dissociated species, so that atomic oxygen partial pressure is around 20% of the stagnation pressure. The mass fraction of dissociated nitrogen is comparable in the two conditions (N images in Fig. 15a,b). The absence of oxygen, whose endothermic dissociation reaction lowers the amount of sensitive enthalpy in the air flow, causes the static temperature to be slightly higher in the case of nitrogen flow. Further details on the effect of the flow conditions on the samples observed behavior will be discussed in Section 4.

Besides the analysis of the fluid flow, the thermal conductivity of the samples at different temperatures was estimated by comparison of the temperature axial profiles measured by the thermo-camera and those calculated by numerical simulations, according to the procedure described in Section 2.4.1. The material conductivity was estimated to

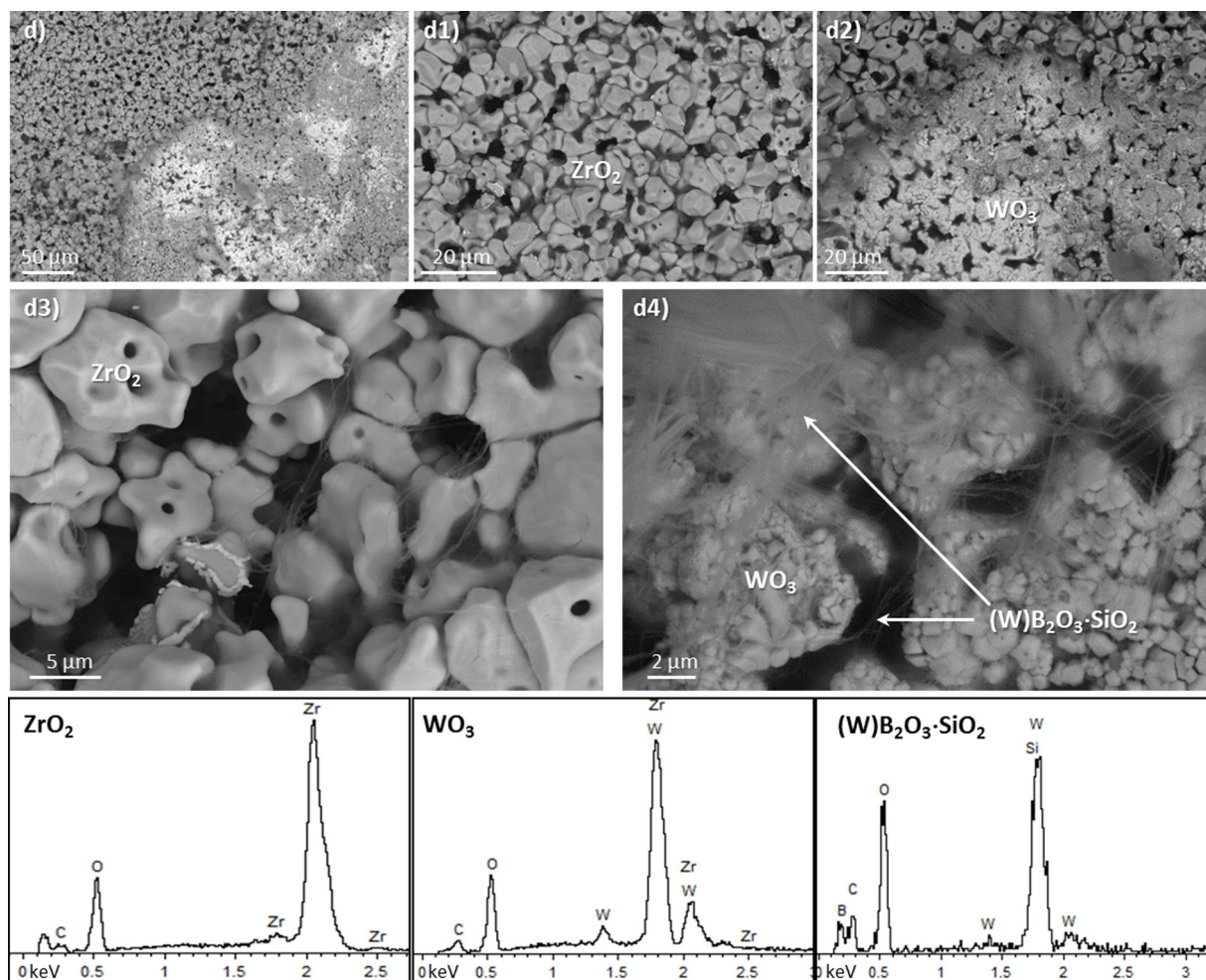


Fig. 8. SEM images of the sub-surface microstructure of sample 1 after the test. d), d1), d2), d3) and d4) are magnifications of zone (d) in Fig. 7.

be around 70 W/(m K) in a temperature range between 1680 and 1850 K, which appears consistent with available literature for ZrB₂-based composites [38]. As an example, the comparison between experimental and numerical axial profiles of temperature for the two samples, at step 4a and 4n respectively, is shown in Fig. 16.

4. Discussion

4.1. UHTC behavior in air – the atoll formation

After test, sample 1 exhibited a relatively smooth surface. Both thermo-camera and SEM of the microstructure indicate that overheating occurred starting from the edges, which triggered oxidation in the sub-scales inducing gas effervescence and consequent outer oxide uplift. In other words, the swollen oxide edges formed a sort of atoll which confined the liquid glass in the center and prevented it from being splashed out. The edge overheating, confirmed also by the thermal camera in Fig. 3a, could be due to the sharp edges of the specimen, which lead to a higher heat flux on the periphery on the sample, as demonstrated by numerical simulations in Fig. 17a, which refers to a cold-wall fully catalytic condition at $H_0 = 20$ MJ/kg in air. Furthermore, silica glass removal could have been favored on the sample periphery by the higher values of shear stress (Fig. 17b), due to the fact that the flow, impinging on the center of the sample (stagnation point), finds its way around the sample accelerating over the surface and attaining maximum velocity around the sample edges. Finally, the higher value of the pressure on the sample center could have suppressed the concurrent outgassing of vapor phases formed in the subscales,

which was instead available only at the edges due to more favorable pressure conditions, Fig. 17c.

In addition, we observed large amounts of B₂O₃ glass in the sub-scales and the formation of the same solid compounds but in a large variety of morphologies depending on the partial pressure of the gaseous environment where they have grown. This points out how, even in a little space range like few hundreds of μm, the developed vapors, i.e. B_xO_y, WO_x, SiO, strongly change the local chemical atmosphere and partial pressure.

In particular, at the basis of the major larger bubbles, accumulation of W/WO₃ was systematically observed, like in sample 2, see Figs. 11b and 13f. This could be due to the fast solidification of the vapors contained in the bubble, suggesting W-compounds to constitute a large fraction of the gas phase and thus to be the major responsible of the overall architecture development and scales uplift. This is in agreement to what has been recently established through volatility diagrams by Zou et al. [7], who assessed that the active oxidation of SiC, and thus SiO formation, is suppressed by the more favorable oxidation of WB into WO₃ and B_xO_y.

4.2. UHTC behavior in nitrogen – the volcano eruption

In order to understand the microstructure evolution of sample 2 and the oxidation mechanism of this system, we can think about a volcano eruption, Fig. 18. Volcanoes are made of liquid, magma, and gas, mainly aqueous vapor, CO/CO₂, SO₂ and others. When liquids and gas find a passage to flow inside the main conduit, they get super-heated, evaporate and become an explosive gas mixture trapped under the

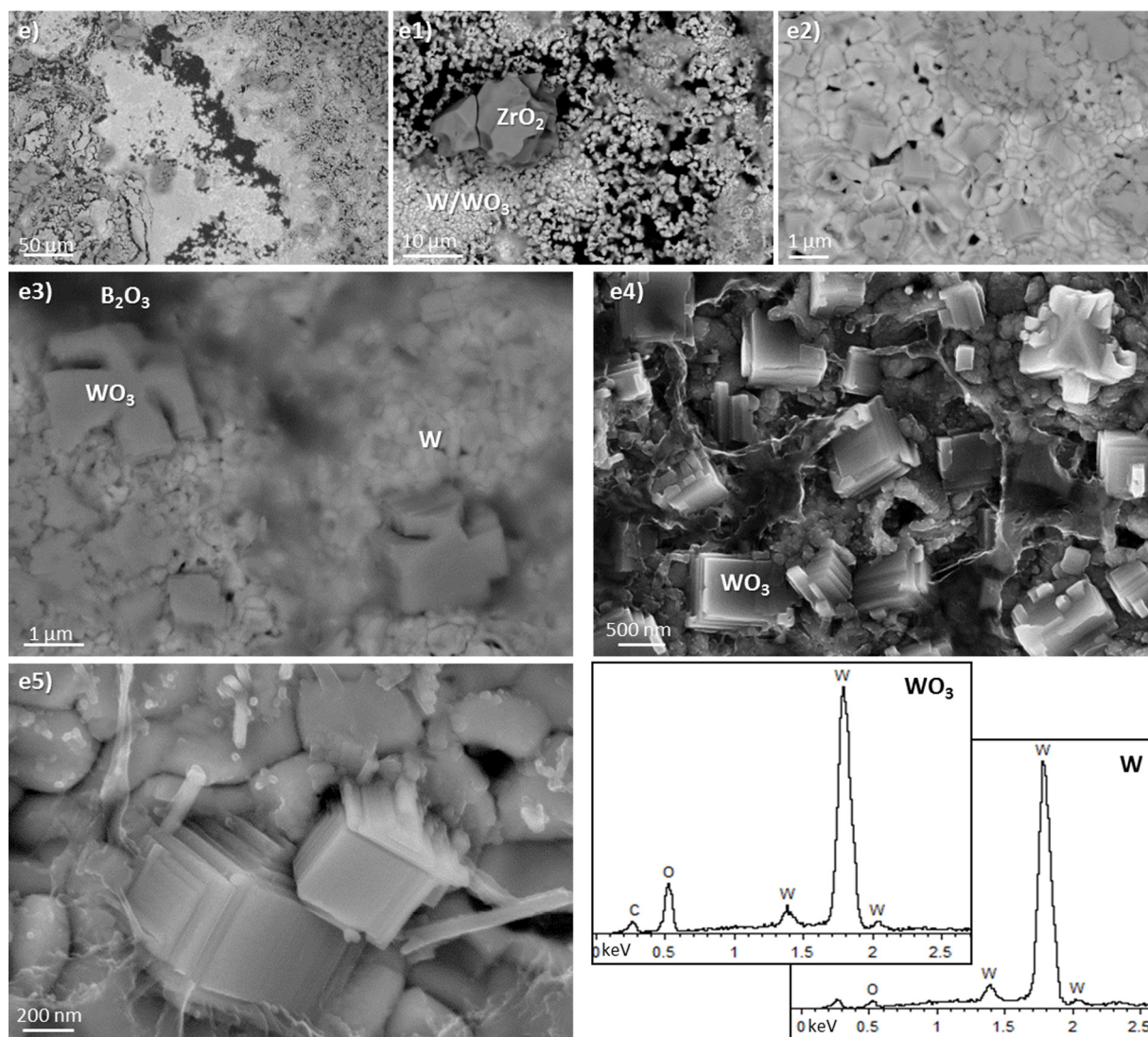


Fig. 9. SEM images of the sub-surface microstructure of sample 1 after the test. e), e1)-e5) are magnifications of zone (e) in Fig. 7 recorded in BSE or Inlens mode. EDS spectra of W and WO₃ are reported.

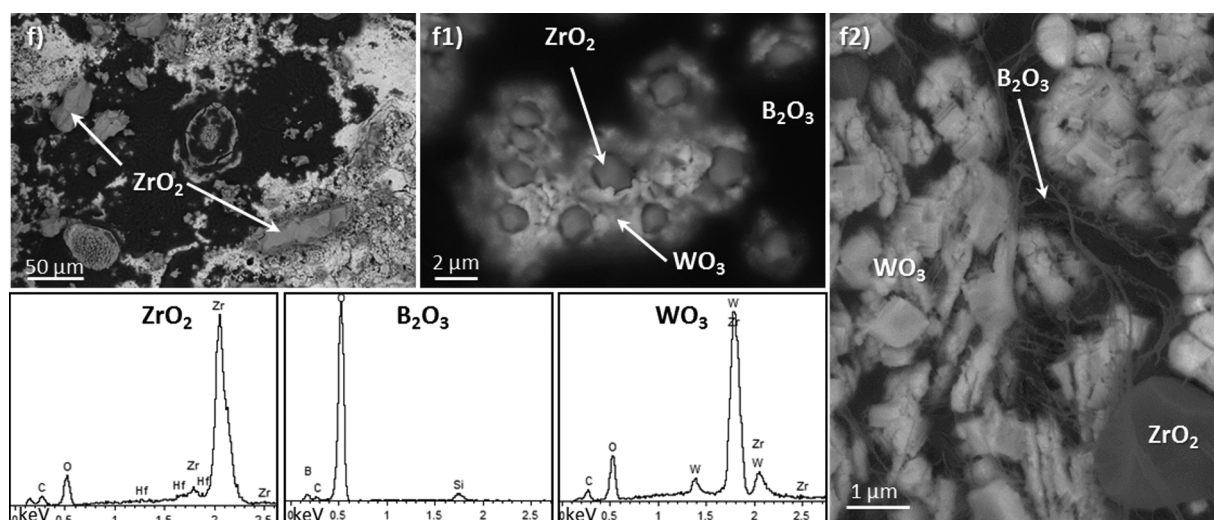


Fig. 10. SEM images of the sub-surface microstructure of sample 1 after the test. Magnification of zone (f) in Fig. 7 with corresponding EDS spectra.

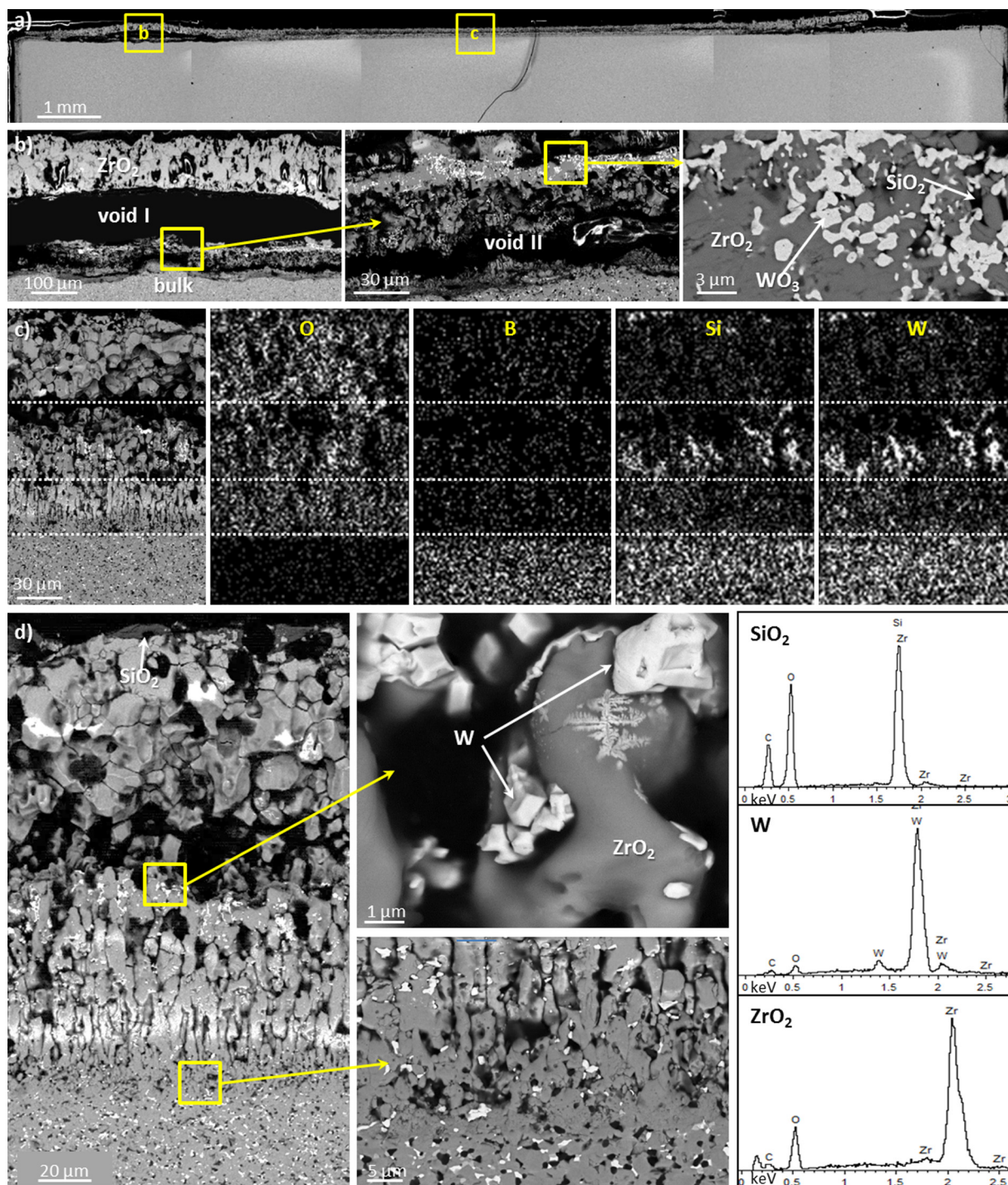


Fig. 11. SEM images of the cross section of sample 1. a) Whole button section showing oxide uplift at the edges, b)-c) magnifications of the microstructure as tagged, c) elemental mapping of the central zone, d) microstructure details in the center with corresponding EDS spectra.

outermost solid scale. Pressure builds up inside the volcano and when reaches a sufficiently high level, it ruptures the ground and the lava is expelled.

In our case, the “ground” is the outermost ZrO_2 -glass scale, the “magma” is the liquid boro-silicate with ZrO_2 and WO_3 , either dissolved or as tiny solid precipitates, and the gas is a mixture of B_xO_y , WO_x , SiO and CO_x . As liquid collection proceeds, the generated gas increases and builds up the pressure inside until the “volcano” eventually pops spreading around the crater solid residuals. This process continues until the volcano is fed, that is until the material undergoes oxidation in the core. Initially, exhaled gas bubbles are very small and find it difficult to

rise up through the melt, even though they are less dense, because the buoyancy force of the bubble is outweighed by the viscous forces of the melt holding it back. But as the melt phase moves upwards, gas bubbles expand as the pressure is further reduced, and many bubbles start to coalesce together forming larger collections of gas, making a foam with the melt. Eventually the amount of gas in the foam is much greater than the amount of melt, causing it to fragment into tiny pieces which are brought up with the rising gases.

As proof of the above described mechanism, we observed on the surface of sample 2 exploded craters, glued fragments and stones of ZrO_2 which seem fallen from above, Fig. 12a, globular $ZrO_2/SiO_2/WO_3$

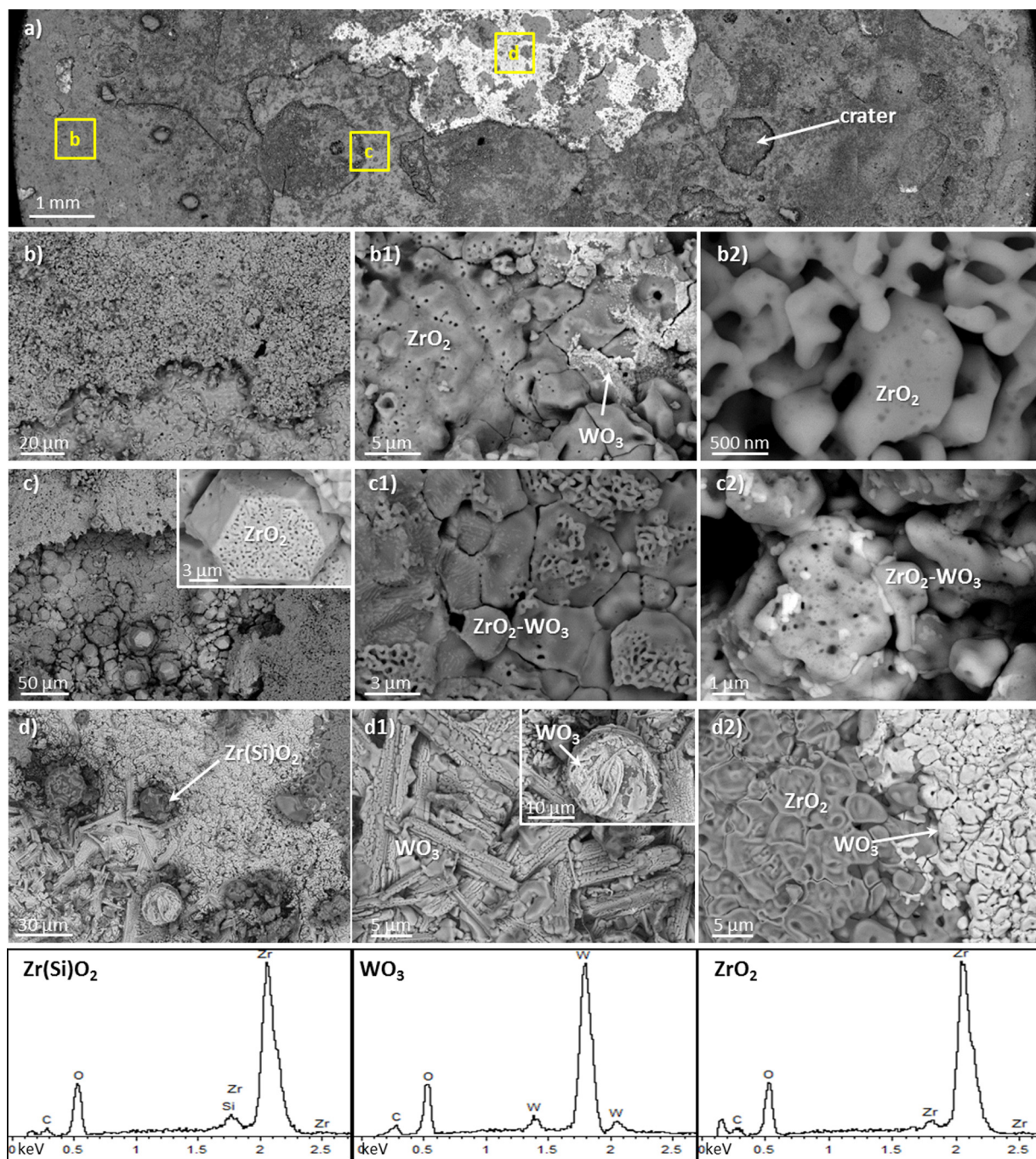


Fig. 12. SEM images of the surface of sample 2 after the test showing a) the overall appearance in the center of the button, b), b1) and b2) microstructure details of the button periphery, c), c1) and c2) of the intermediate zone and d), d1) and d2) of the button center. Below the EDS spectra of the mentioned phases are reported.

aggregates, Fig. 12d, and, as confirmation of the violent gas escape in a sort of sparkling champagne, all zirconia grains are featured by little diffused nano-porosities, Fig. 12c, in all analogous to pumice stone.

Moving to the sub-layers, the reason of multiple cell-like oxide scale development could be related to a cyclic mechanism associated with periodical cracking at confined areas when a dense oxide layer reaches a critical thickness, in analogy with the oxidation of pure tungsten [39]. Alternatively, a sub-surface “eruption” may take place when the external pressure decreases by either cracking of the solid oxide above or by lateral leakage of the melt through micro-cracks which induces a pressure decrease of the remaining fluid.

Another important aspect to consider is the composition of our “magma”, a controlling factor which determines its viscosity. Indeed a

very viscous melt can trap ex-solving gases causing a build-up of pressure, whereas a less viscous one allows gases to escape easily. Melts with a high silica content will be very viscous, whereas melt with a high boron oxide content will be very fluid. Along its climb back towards the surface, the melt meets variation of pressures which can induce preferential crystallization of its components with consequent composition and viscosity variation. A different composition of the melt between sample 1 and sample 2, promoted by oxygen, might be the cause for a milder or stronger eruption, respectively.

4.3. Overall remarks

In light of these experiments simulating the hypersonic

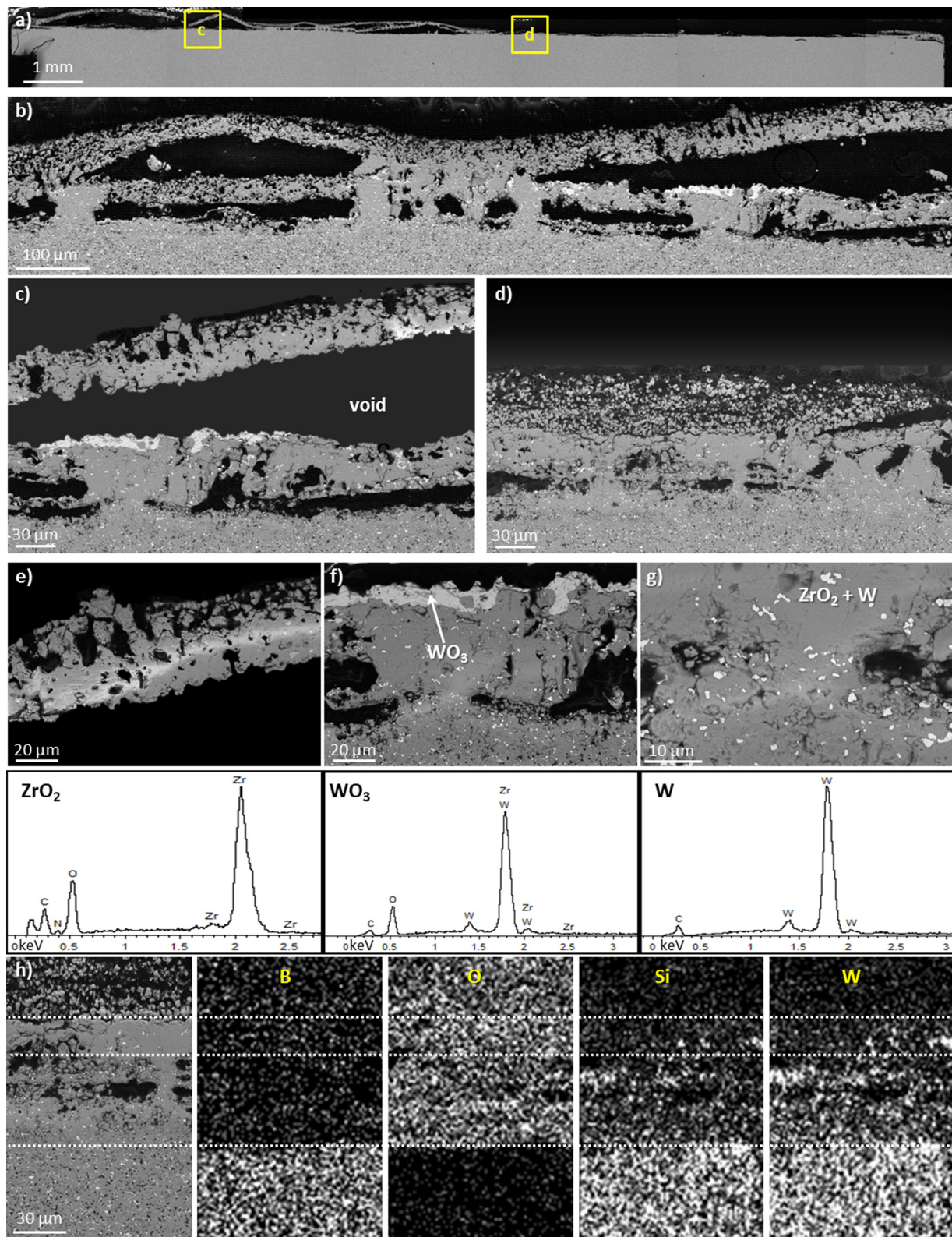


Fig. 13. SEM images of the cross section of sample 2. a) Whole button section showing oxide uplift and b) multiple delamination, c)-d) magnifications of the microstructure with and without bubbles, e-g) microstructural details of c) with corresponding EDS spectra below and g) elemental mapping of zone d).

environment, it seems that the microstructure evolution under air or nitrogen flow is alike, but a larger amount of oxygen in the case of air flow is actually beneficial to boost a healing mechanism by formation of a glassy outermost phase which partially mitigates or retards the material degradation. It is our opinion that the presence of this layer acts

as barrier to the oxygen diffusion from the high-enthalpy gas flow to the bulk material. This is also confirmed by the temperature profiles recorded by the pyrometer, Fig. 2, where the temperature jump occurs at temperature 200 K higher, and in a more gradual fashion, when the button is tested in air. As a matter of fact, the limited amount of oxygen

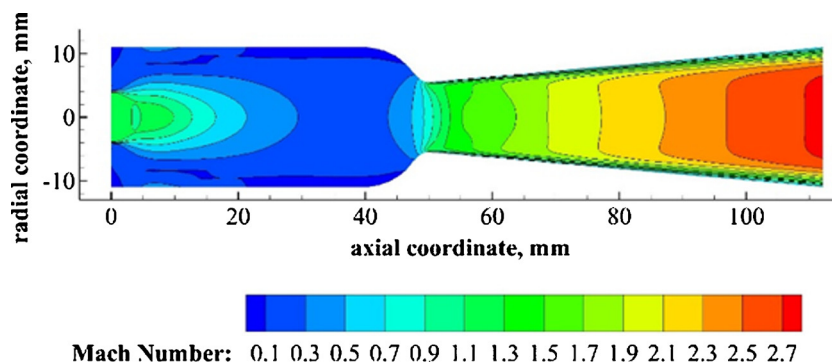


Fig. 14. Mach Number distribution inside mixing chamber and nozzle during test in air, Step 7a.

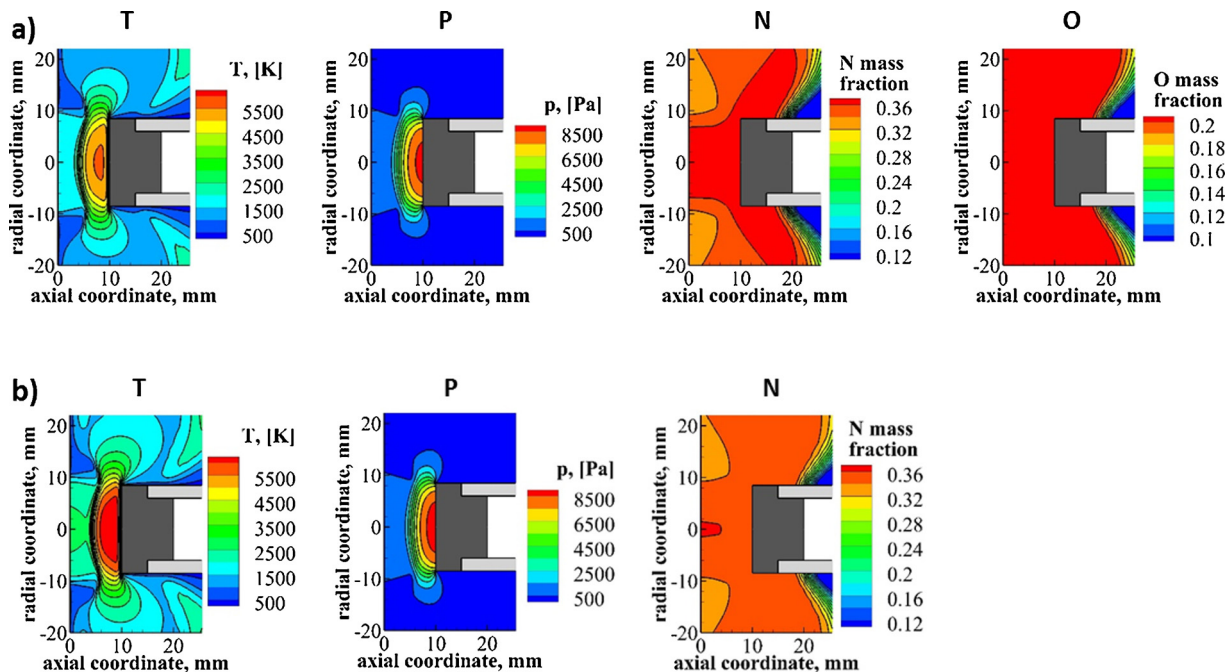


Fig. 15. (T) Temperature, (P) Pressure, (N) N mass fraction and (O) O mass fraction distributions around the test article in a) air flow, Step 7a, and b) nitrogen flow, Step 5n.

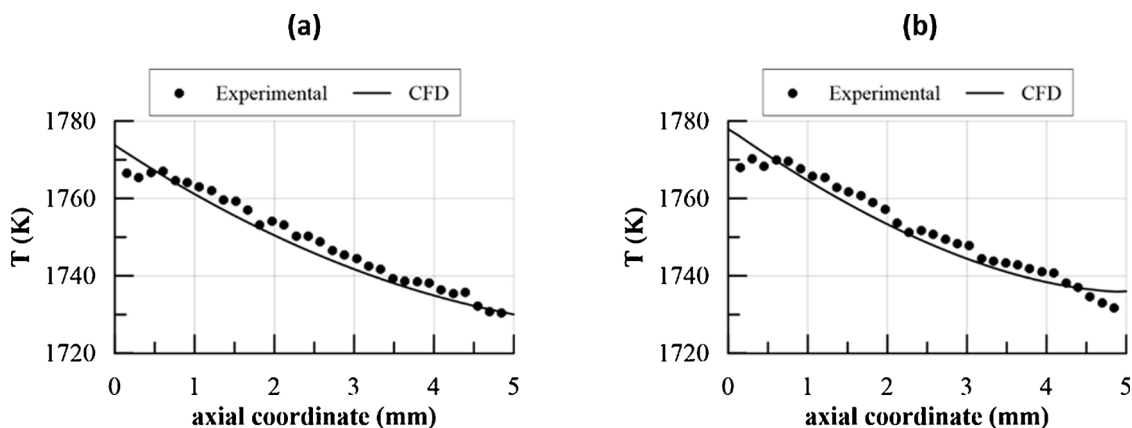


Fig. 16. Comparison between experimental and numerical temperature axial profile ($k = 70 \text{ W/m K}$), for (a) sample tested in air (Step 4a) and (b) sample tested in nitrogen (Step 4n).

in the nitrogen flow, with a partial pressures in the order of tens of Pa, is not able to trigger this healing mechanism by formation of glassy borosilicate. This is responsible for a more violent oxidation process, which has been here associated to a volcano eruption. Due to the unhampered

diffusion of oxygen from the gas flow to the bulk material, the gas oxides generated by evaporation in the sample reach a relatively high partial pressure, until they get super-heated, and become an explosive gas mixture breaking the surface solid scale.

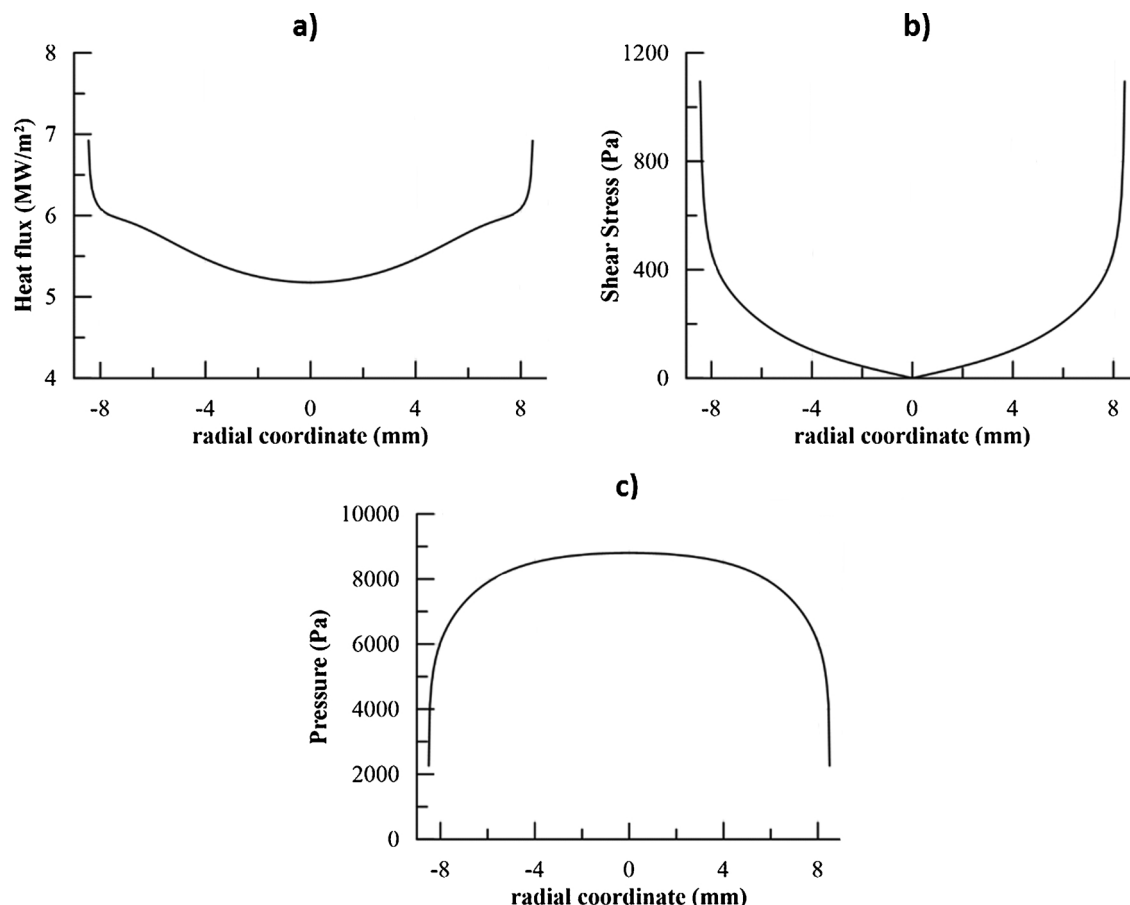


Fig. 17. (a) Cold wall fully catalytic heat flux, (b), shear stress and (c) static pressure profiles over the sample front surface during test in air at $H_0 = 20$ MJ/kg.

Based on the outcomes of the aero-thermo-dynamic testing here presented, we cannot say this material is optimized for extreme environments. One interesting aspect to underline regards the effect of the even relatively low volume fraction of W-compounds on the oxidation behavior of this strong UHTC, which is appreciable looking at the overall architecture of the modified scale, Fig. 13b. Cross section views revealed that, under the outermost external layer, a relatively dense scale with periodic bands of cavities or empty cells, was present. Formation of such oxide structure with multilayered cavities has been previously observed during the oxidation of pure W already starting from 600–700 °C [39,40]. The formation of such peculiar architecture was ascribed to continuous formation and cracking of the oxide scale, Fig. S2. What is important to underline is that the cross section appearance strongly suggests that the development of such subsurface architecture is affected by just 5 vol% of W-compounds.

One approach to improve the material performance could be to further decrease the starting WC amount so that almost no residual WB particles are present at the boride grain boundaries, but W is present just finely dispersed in the $(\text{Zr}, \text{W})\text{B}_2$ solid solution [41]. This would guarantee the high temperature mechanical performances and surely decrease the entity of the bursting phenomena in the oxide scale and hence improve its structural stability.

5. Conclusions

A ceramic classified as “super-strong” at temperature exceeding 2000 °C and based on ZrB_2 -WC-SiC was characterized in supersonic arc-jet wind tunnel to study the effect of different flow composition: simulated air or pure nitrogen, i.e. with oxygen partial pressures of thousands of Pa for the former and tens of Pa for the latter. The sintered

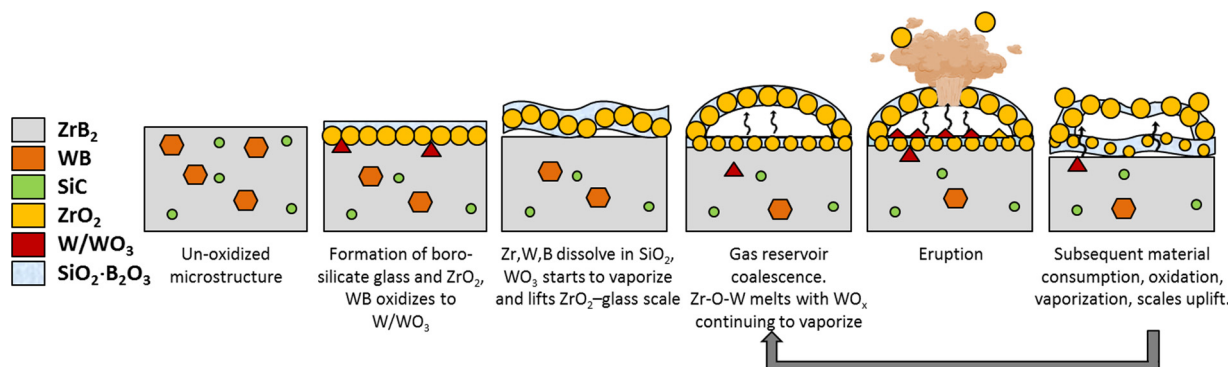


Fig. 18. Sketch of the oxidation mechanism of the ZrB_2 -WC-SiC ceramic upon exposure to supersonic flow.

ceramic was composed of $\text{ZrB}_2/(\text{Zr,W})\text{B}_2$ grains with scattered SiC and WB particles. The maximum temperature achieved on the surface of the specimens with flat button geometry was 2650 and 2800 K, at specific total enthalpies over 20 MJ/kg. Surface instabilities, related to formation and bursting of bubbles and temperature jumps of 500–600 K were detected by infrared pyrometer and thermo-camera at high enthalpies. These phenomena were then associated to the microstructural features observed by SEM-EDS on the surface and cross-section.

The comparison of the two test specimens revealed analogous oxide architectures, generally characterized by WO_x -based vapors in the sub-scales, with particular effervescence in the case of the sample exposed to nitrogen flow. In the sample exposed to air, formation of surface silica-based glass mitigated bursting phenomena. The solid crystalline compounds, ZrO_2 , W and WO_3 were present in a wide spectrum of morphologies, testifying large partial pressure variation from point-to-point. A volcano-model was proposed to explain the oxidation mechanism of this system in the absence of a protecting glassy silica layer, involving the cyclic accumulation of gases at high pressure in the sub-scales and eruption of fluids.

Two major conclusions were drawn. First, it seems that presence of oxygen in the incoming flow is beneficial for retarding the material consumption, thanks to the activation of partially healing mechanism by SiC oxidation and B_2O_3 retention into a protective silica glass.

Second, it seems that introduction of W-compounds in the present concentration, 5 vol%, although fundamental for ZrB_2 strength retention above 2000 °C, dictates the formation of the sub-surface architecture making it resemble to that of pure W. This crucial observation leads to propose that the volume fraction of W-compounds should be well below this value in order to improve oxidation resistance in hypersonic environments.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time due to legal or ethical reasons.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.corsci.2019.108125>.

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